



BUILDING PERMITS CLUSTERING ANALYSIS

Discovering Natural Groups of Construction Projects in San Francisco

AGENDA

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Project Goal & Motivation

Why cluster building permits?

02

Dataset & Feature Selection

What we used and what we deliberately excluded

03

Methodology

Preprocessing, choosing k, final model

04

Cluster Results & Profiles

Three distinct groups of projects

05

Neighborhood Analysis

Where each type of project concentrates

06

Insights, Code & Next Steps

Key findings and future work

PROJECT GOAL

Group building permits by the nature of the project (size, cost, construction type) then discover which neighborhoods are associated with each group.

Unsupervised Discovery

Let the data reveal natural groups instead of imposing labels

Actionable Insights

Understand where large vs small projects concentrate in the city

Policy & Planning Value

Help city planners anticipate workload by neighborhood type

DATASET OVERVIEW

~111k

Rows after cleaning

k = 3

Final clusters

SF

City data source

Cost > \$1k

Filter applied

Features Used for Clustering

- ▶ Number of Stories (Existing & Proposed)
- ▶ Plansets
- ▶ Construction Type codes
- ▶ Existing Use & Proposed Use
- ▶ Number of Units (Existing & Proposed)
- ▶ Estimated Cost & Revised Cost
- ▶ Permit Type Definition
- ▶ Construction Type Descriptions

KEY DESIGN DECISION

Neighborhood was deliberately EXCLUDED from clustering features

We want the algorithm to group permits by the nature of the project (size, cost, type of work). Afterwards we examine which neighborhoods appear in each group.

Avoid wrong reasoning

If we include neighborhood, clusters would largely just recreate geographic areas.

Discover project types first

Clustering on physical & cost attributes reveals the real structure of the work.

Then map to location

After clustering we can ask: where do mega-projects vs everyday permits concentrate?

METHODOLOGY

1

Clean & Filter

Keep Revised Cost > \$1,000.
Median / Unknown imputation.

2

Select Features

Size, cost, construction type, permit type, use. Exclude neighborhood.

3

Preprocess

StandardScaler for numerics +
OneHotEncoder for categoricals.

4

Choose k

Elbow method + Silhouette score on
a sample.

5

Final Model

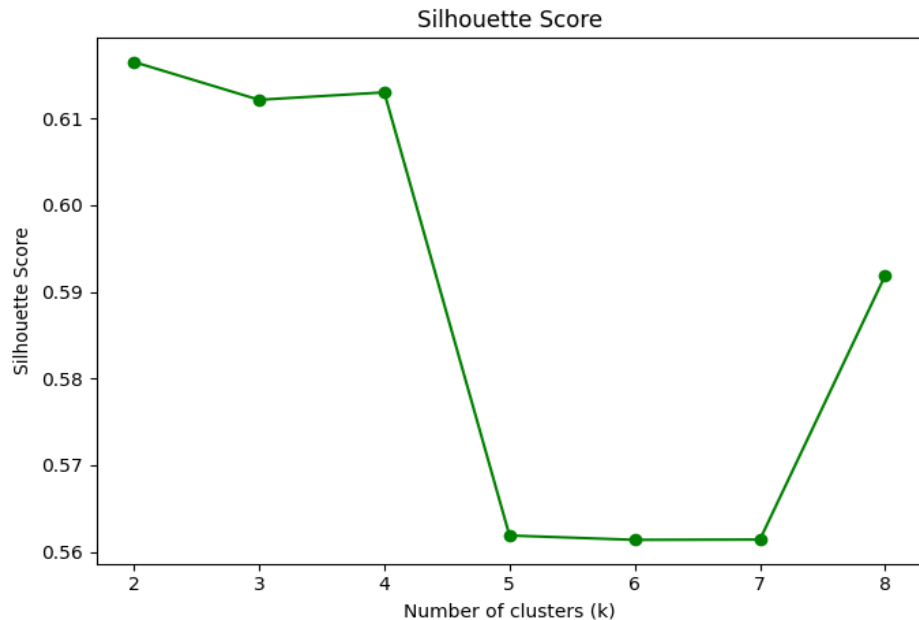
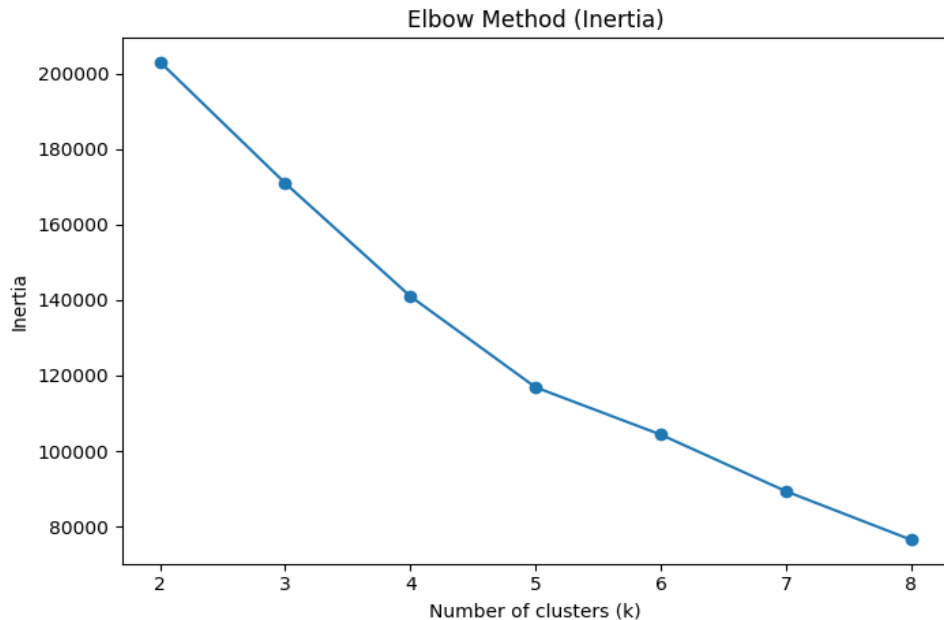
KMeans with chosen k on the full
dataset.

6

Interpret

Profiles + neighborhood distribution
+ PCA plot.

CHOOSING THE NUMBER OF CLUSTERS (k)



We selected $k = 3$ for the final model. It provides a clear separation between everyday permits, significant mid/high-rise projects, and rare mega high-rise developments.

FINAL CLUSTERS

CLUSTER 0

Everyday / Residential

91,932

permits

~83% of total

CLUSTER 1

Mid-to-High Rise

17,852

permits

~16% of total

CLUSTER 2

Mega High-Rise

1,295

permits

~1% of total

CLUSTER 0

Everyday / Residential Permits

91,932 (~83%)

Size

\$59,400

Avg Revised Cost

2.7

Avg Stories

6.1

Avg Units

Description

The dominant group of ordinary residential and small scale work. Typical projects include home renovations, minor additions, and low rise multi unit buildings. Costs and building sizes stay low/modest.

Top Neighborhoods

Sunset/Parkside (8.1%) • Mission (7.6%) • West of Twin Peaks (6.9%) • Outer Richmond (5.3%) • Noe Valley (5.0%)

Classic residential neighborhoods, exactly where everyday permits are expected to concentrate.

CLUSTER 1

Significant Mid-to-High Rise Projects

17,852 (~16%)

Size

\$212,800

Avg Revised Cost

17.6

Avg Stories

27.1

Avg Units

Description

Substantial commercial and residential projects that are clearly larger than everyday work. Typical mid-to-high-rise buildings with meaningful unit counts and costs in the low hundreds of thousands.

Top Neighborhoods

Financial District / South Beach (56.7%) • South of Market (9.6%) • Chinatown (5.5%) • Western Addition (4.3%) • Nob Hill (4.1%)

Strongly concentrated in the downtown core and similar dense neighborhoods.

CLUSTER 2

Mega High-Rise Developments

1,295 (~1%)

Size

\$248,500

Avg Revised Cost

26.8

Avg Stories

521

Avg Units

Description

Rare but extreme projects: very tall buildings with hundreds of units. Although average cost is lower than expected for this scale, the unit count and height clearly identify them as major high-rise developments.

Top Neighborhoods

Financial District / South Beach (36.8%) • Tenderloin (30.2%) • Mission Bay (8.7%) • Western Addition (5.6%) • Japantown (5.6%)

Concentrated in the downtown core and areas of major dense public.

NEIGHBORHOOD PATTERNS SUMMARY

Residential Neighborhoods → Cluster 0

Sunset/Parkside, Mission, West of Twin Peaks, Outer Richmond, and Noe Valley are dominated by everyday low-rise permits.

Downtown Core → Clusters 1 & 2

Financial District / South Beach is the clear hotspot (57% of Cluster 1, 37% of Cluster 2). SoMa, Chinatown, and Nob Hill also appear.

Tenderloin & Mission Bay → Cluster 2

These areas show elevated presence of the rare mega high-rise projects, reflecting dense public and major development activity.

CLUSTERS IN 2D (PCA VISUALIZATION)



PCA reduces the high-dimensional feature space so we can visually confirm that the three groups are reasonably separated.

KEY INSIGHTS

Clear Scale Separation

K-Means separated permits into everyday work (~83%), significant mid/high-rise (~16%), and rare mega high-rise (~1%) based on size and cost attributes.

Geography Follows Project Type

Large projects concentrate heavily in the Financial District / South Beach and Tenderloin. Everyday permits dominate residential neighborhoods.

Design Choice

Excluding neighborhood from the features prevented the model from simply rediscovering geography and forced it to learn project characteristics.

Actionable for Planning

City planners can anticipate that a surge in Cluster 1 or 2 permits will heavily impact downtown infrastructure and review resources.

KEY CODE SNIPPETS

Feature selection (neighborhood excluded)

```
features = ["Number of Existing Stories", "Number of Proposed Stories",  
            "Existing Units", "Proposed Units", "Plansets", "Revised Cost",  
            "Permit Type Definition", "Existing Use", "Proposed Use", ...]  
# Neighborhood deliberately left out
```

Preprocessing + KMeans Pipeline

```
pipeline = Pipeline([  
    ("preprocessor", ColumnTransformer([...])),  
    ("cluster", KMeans(n_clusters=3, random_state=42, n_init=10))  
])  
pipeline.fit(X)
```

Neighborhood analysis after clustering

```
ct = pd.crosstab(df_cluster["Cluster"],  
                 df_cluster["Neighborhoods - Analysis Boundaries"], normalize="index")
```

THANK YOU

Questions & Discussion

Three clear project types were found:
Everyday residential • Mid/High-rise • Mega high-rise

Geography follows the nature of the work.